

MetricSpace: Learning Better Representations of Clinical Trial Eligibility Criteria

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1 Read This First

Before starting, each team member should read the original AutoCriteria paper [1] and understand its basic workflow: an LLM reads free-text eligibility criteria from a clinical trial and converts them into a more structured format. The paper is the starting point for this project.

Team members should then review the Modern AutoCrit repository [2]:

<https://github.com/JonathanMa03/modern-autocrit/tree/main>

The repository contains the working system that this project will extend. Teammates do not need to understand every implementation detail before contributing, but they should be able to run the application, inspect one completed extraction, and follow a criterion from source protocol text to its structured output.

Modern AutoCrit is our expanded implementation of that idea. The original AutoCriteria work established the protocol-to-structured-criteria concept. Modern AutoCrit adds the project infrastructure needed for MetricSpace:

- retrieval of ClinicalTrials.gov records by NCT identifier;
- parsing of ClinicalTrials.gov API v2 records;
- atomic segmentation of compound eligibility text into separately comparable criteria;
- structured entity–attribute–value representations;
- normalized values, intervals, operators, and units;
- governed clinical terminology and reusable attribute identifiers;
- validation, reconciliation, confidence, and human-review outputs;
- recoverable jobs and persistent intermediate results;
- modular backend code rather than one long extraction script; and
- a browser interface for running extraction and inspecting results.

The features most relevant to MetricSpace are atomic segmentation, normalized structured values, terminology identifiers, source evidence, confidence, and reviewer decisions. They provide the inputs and quality signals needed to construct criterion pairs. The browser interface and job-recovery layer are useful engineering support, but they are not the research contribution of MetricSpace.

You do not need to understand the entire codebase immediately. The important input and output are:

Protocol eligibility text $\rightarrow$ atomic, structured eligibility criteria

For example, Modern AutoCrit should turn “participants must be at least 18 years old” into a record containing the concept *age*, the operator $\geq$, the value 18, and the unit years.

This project begins after extraction. We want to compare criteria from different protocols and train an embedding model that understands clinically important differences.

2 The Project in Plain Language

Current sentence-embedding models often reward similar wording. That is not always clinically correct:

- “age ≥ 18 ” and “age < 18 ” look almost identical but describe different groups;
- “no cardiovascular disease” and “controlled hypertension is allowed” discuss the same topic but are not equivalent;
- “creatinine clearance ≥ 30 mL/min” and “eGFR ≥ 30 mL/min/1.73 m²” may look similar but use different measurements; and
- two differently worded criteria may actually mean the same thing.

Our goal is to build a new dense embedding model for eligibility criteria. Criteria with similar clinical meaning should be close in the learned space. Criteria that only look similar, but differ because of negation, operators, thresholds, timing, or exceptions, should be separated.

The system will use an agent to find difficult examples, ask an LLM for an initial judgment, run rule-based checks, send uncertain cases to humans, and turn accepted labels into model-training data.

One-sentence aim

Build and evaluate an agentic workflow that creates human-validated criterion pairs from clinical protocols and uses them to fine-tune a clinically meaningful dense embedding model.

3 What We Are Trying to Learn

For each pair of criteria, we will assign one of the following relationships:

- equivalent;
- partially overlapping or compatible;
- criterion A is more restrictive;
- criterion B is more restrictive;
- contradictory or effectively disjoint;
- related topic, but not directly comparable;
- unrelated; or
- too ambiguous to decide.

The main model has two jobs:

1. **Retrieval:** given one criterion, retrieve other criteria with comparable clinical meaning.
2. **Relationship prediction:** explain whether a retrieved pair is equivalent, overlapping, directional, contradictory, or unrelated.

These are separate because ordinary embedding distance is symmetric, while “A is more restrictive than B” is directional.

4 Data We Need

4.1 Protocol corpus

Clinical protocols are the main source material. We will begin with one or two therapeutic areas so that the trials are reasonably comparable. For every protocol, we will keep:

- the original eligibility text;
- individual criteria extracted by Modern AutoCrit;
- structured values, operators, units, negation, timing, and other qualifiers;
- the source protocol and location; and
- basic trial metadata.

The dataset must include more than age. We want examples involving demographics, laboratory thresholds, disease stage, biomarkers, prior treatment, medication use, comorbidities, performance status, reproductive requirements, timing, severity, negation, and exceptions.

Chia is a useful example of a large structured eligibility-criteria dataset [3]. Prior work has also studied embeddings for eligibility clustering and trial similarity [4, 5]. We still need our own pair labels because those resources do not directly answer our relationship question.

4.2 How we find useful pairs

First, we will embed all criteria using existing frozen models. We will then inspect nearest neighbors and deliberately collect cases where conventional embeddings fail. Useful cases include:

- opposite operators or negation despite high similarity;
- changed values, units, or time windows;
- equivalent meaning written differently;
- broad rules paired with narrower exceptions;
- similar vocabulary used for different clinical purposes; and
- uncertain or poorly extracted criteria.

We will also include random and clearly unrelated pairs. Otherwise, a benchmark made only from difficult failures would not reflect normal model behavior.

Training, validation, and test sets will be separated by protocol. Criteria from the same source or template must not appear on both sides of the split.

5 How Labels Become Training Data

The workflow is:

Protocol pair $\rightarrow$ agent tools $\rightarrow$ LLM suggestion $\rightarrow$ human review $\rightarrow$ accepted label $\rightarrow$ model training.

5.1 LLM suggestion

The LLM receives the two criteria, their structured fields, and enough protocol context to interpret them. It proposes a relationship, confidence, and short rationale. This proposal is not treated as truth.

Automatic checks reject obvious mistakes. For example, a label should reverse correctly when A and B are swapped, incompatible units should be flagged, and exact disjoint intervals cannot be called overlapping.

5.2 Human review and steering

Reviewers accept, correct, reject, or mark each proposed label as ambiguous. Early examples should be independently reviewed by two people so that we can improve the label guide and measure agreement. Only accepted or adjudicated labels enter training.

As the model improves, the agent will prioritize examples where:

- the model is uncertain;
- the LLM and structured tools disagree;
- input order changes the answer;
- the example is unlike the existing training data; or
- reviewers frequently disagree.

This is the human-steering loop: human corrections guide which examples are collected next and how the model is retrained.

6 The Dense Embedding Model

We will start from a pretrained text encoder rather than train a language model from scratch. We will then fine-tune it on our human-validated pairs. The contribution is the new protocol-derived dataset, clinically difficult negatives, training setup, agentic labeling process, and evaluation.

Training examples have three broad roles:

- **Positive pairs:** equivalent or clinically interchangeable criteria;
- **ordinary negatives:** unrelated criteria; and
- **hard negatives:** criteria with similar wording but an important clinical difference.

The encoder $f_\theta(c)$ maps criterion c to a vector. Its basic distance is

$$d_\theta(A, B) = 1 - \cos\{f_\theta(A), f_\theta(B)\}.$$

Contrastive training pulls positive pairs closer and pushes negative pairs apart. Hard negatives teach the model that an operator, number, negation, time window, or exception can matter even when the rest of the sentence is unchanged.

A small classification head will use both criterion vectors to predict the explicit relationship label. This head handles directional relationships that cosine distance alone cannot represent.

Structured rules are used for labeling and quality control. We will not average structured, embedding, population, and LLM scores into one score with unclear clinical meaning.

7 Population Information from Manuscripts

We probably will not have individual patient data. Therefore, we will not claim to calculate how many individual patients would change eligibility between two trials, and we will not simulate an unknown patient population as if it were observed.

Instead, we may link protocols to their trial manuscripts and collect reported baseline characteristics from Table 1 or supplementary tables. Examples include age, sex, race or ethnicity, disease stage, performance status, prior treatment, and disease-specific severity measures.

For a characteristic reported by both trials, we can calculate a standardized difference. For continuous variables:

$$\Delta_k(a, b) = \frac{\bar{x}_{ak} - \bar{x}_{bk}}{\sqrt{(s_{ak}^2 + s_{bk}^2)/2}}.$$

Binary proportions can be standardized in a similar way. The main output should be a table or profile showing which reported characteristics differ, not a single unexplained population score.

This is a secondary analysis. It asks whether protocols that look similar according to their eligibility criteria also report similar enrolled populations. It does not establish causality or transportability. Published summaries omit information and cannot recover the full joint patient distribution. Population-adjustment literature explains these limitations [6, 7].

Before implementing this analysis, we need a short literature review and a common dictionary for baseline variables. If too few manuscripts report comparable fields, this component should be reduced rather than filled with unsupported assumptions.

8 What Makes the System Agentic

The agent coordinates a workflow rather than simply generating an answer. It will:

1. select candidate criterion pairs;
2. retrieve the original protocol context;
3. choose structured, terminology, embedding, or LLM tools;
4. check units, operators, negation, and input order;
5. send uncertain cases to human reviewers;
6. record the final label and its provenance;
7. choose new examples using uncertainty and disagreement; and
8. apply the fine-tuned model with an audit trail.

The agent does not change model weights during a comparison and does not invent a distance. Training happens offline on reviewed data. At runtime, the agent loads a versioned model and decides whether a result is reliable enough to accept.

9 How We Will Evaluate It

9.1 Models to compare

- TF-IDF or another lexical baseline;
- frozen general sentence encoder;
- frozen biomedical sentence encoder;
- structured rules where applicable;
- LLM judgment without fine-tuning;
- our fine-tuned dense encoder; and
- the full agent workflow with review and abstention.

We will also remove important pieces one at a time, such as hard negatives, human corrections, or critic review, to see what actually improves performance.

9.2 Measures

For retrieval, we will report $\text{recall}@k$, mean reciprocal rank, and ranking quality. For relationship prediction, we will report macro-F1, class-level errors, confusion matrices, probability calibration, and input-order consistency. Confidence intervals will resample protocols rather than individual pairs because pairs from one protocol are related.

For the agent, we will measure how often its proposed labels are accepted or corrected, whether it catches conflicts, whether it escalates the right cases, how much reviewer time is needed, and how model performance changes after each annotation round.

Age remains a useful unit test for boundaries and operators. It is not the main benchmark.

10 Team Responsibilities

Owner	Main job	Deliverables
Member A: data and protocols	Build the protocol/manuscript corpus, run Modern AutoCrit, organize criterion pairs, and lead annotation.	Clean criterion dataset, provenance, annotation guide, pair labels, data splits, reviewer agreement, and baseline-characteristic table.
Member B: embeddings and statistics	Run baseline encoders, fine-tune the new encoder, build the relationship classifier, and perform statistical evaluation.	Baseline results, training pipeline, trained model, calibration, confidence intervals, ablations, and population-summary analysis.

Owner	Main job	Deliverables
Member C: agent and application	Build the agent workflow, tools, checks, LLM labeler, critic, review routing, and demonstration.	Working agent, tool interfaces, audit logs, active-learning loop, escalation results, and end-to-end application.

All three members will read the AutoCriteria paper, help label difficult examples, review one another’s work, run integration tests, interpret errors, and prepare the presentations and report.

11 Timeline

Dates	Goal
October 3–6	Everyone reads AutoCriteria. Agree on relationship labels, therapeutic areas, protocol selection, annotation format, and software interfaces.
October 7–12	Run Modern AutoCrit on the first protocols. Test frozen embedding models. Retrieve nearest neighbors and collect clear failure examples. Pilot the LLM labeler and human-review form.
October 13–16	Label a small multi-domain dataset, measure initial reviewer agreement, build the first training dataset, and implement basic agent checks.
October 17–19	Produce preliminary retrieval results, failure examples, a working agent trace, and, if enough labels exist, a small fine-tuning experiment. Prepare the presentation.
October 20	Progress presentation: explain AutoCriteria and Modern AutoCrit; describe the corpus; show embedding failures; explain the LLM-to-human labeling loop; report baseline results; demonstrate one agent run; present risks and the remaining plan.
October 21–November 2	Expand protocols and labels, improve prompts and checks, finish double review, collect manuscript summaries, and freeze train/validation/test splits.
November 3–16	Fine-tune the encoder, build the relationship classifier, calibrate it, and run the main ablations.
November 17–30	Run held-out tests, agent and abstention experiments, manuscript population analysis, robustness checks, and error analysis.
Final period	Freeze data and models, reproduce results, finish the application, write the report, and prepare the final presentation.

12 TODO List

Before October 20

- ☐ Read the AutoCriteria paper and review the Modern AutoCrit input/output.
- ☐ Choose one or two therapeutic areas and define which protocols qualify.

- ☐ Agree on relationship labels and write short examples for each label.
- ☐ Run Modern AutoCrit on the first protocol batch and inspect its output.
- ☐ Select and record at least two frozen embedding baselines.
- ☐ Retrieve nearest neighbors and collect multi-domain failure examples.
- ☐ Build the LLM label format, rule checks, and human-review form.
- ☐ Double-label a pilot dataset and calculate agreement.
- ☐ Split data by protocol and check for leakage.
- ☐ Create a minimal fine-tuning dataset and training script.
- ☐ Demonstrate one full agent and review workflow.
- ☐ Prepare the October 20 slides, figures, and risk list.

After October 20

- ☐ Expand labels using uncertainty and tool disagreement.
- ☐ Define positive, ordinary-negative, and hard-negative sampling.
- ☐ Fine-tune and calibrate the dense encoder.
- ☐ Train the relationship classifier.
- ☐ Test the contribution of hard negatives, LLM suggestions, human corrections, and the critic.
- ☐ Review aggregate population-comparison literature and define allowed claims.
- ☐ Extract comparable manuscript baseline characteristics with provenance.
- ☐ Evaluate retrieval, relationship classification, calibration, and agent routing.
- ☐ Complete domain-specific and out-of-distribution error analysis.
- ☐ Freeze code, prompts, data, model versions, and reproducibility instructions.

13 Scope Boundaries

The minimum project is a multi-domain protocol corpus, two frozen baselines, a reviewed pair dataset, a fine-tuned dense encoder, a relationship classifier, an agentic labeling and review loop, and held-out evaluation.

The manuscript baseline analysis is included only if enough comparable variables are available. We will not claim individual-level population disagreement, causal transportability, or simulated patient outcomes without suitable data. We will also avoid an unexplained weighted score that mixes unrelated quantities.

References

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